Questions about 1D array :

Q1/ write java program to illustrate creating an array of type double and print each element with its index

```
public class ArrayExam {
        public static void main(String[] args) {
        double[] array= {22.4, 13.3, -6.5, 9};
        for (int c = 0; c < array.length; c++)
        System.out.println("Element at index "+ c + " = " + array[c]);
        }
}
```

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Q2/ Write a Java program to find the summation of the elements in an array then print out the square and square root of that summation.

```
public class ArrayExampl {
        public static void main(String[] args) {
        int numbers[] = {8, 2, 6, 4, 3};
        int sum = 0;
        for (int i=0; i < numbers.length; i++)
                sum = sum + numbers[i];
        System.out.println("The summation is: " + sum);
        System.out.println("The square root is: " +Math.sqrt(sum));
        System.out.println("The square is: " +(sum*sum));
        }
```

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Q3/ Write java program to create 1d array has 10 float numbers performce the following:
1- sum all items from index 0 to 3
2- find square all items from 3 to end of array

```
public class ArrayEx {
        public static void main(String[] args) {
        float summation = 0;
        float arraya[] = {0.1f , 0.2f , 0.3f , 0.4f , 0.5f , 0.6f , 0.7f , 0.8f , 0.9f ,1.0f};
        for (int i = 0 ; i < 4 ; i++)
                summation = summation + arraya[i];

        System.out.println("The summation is "+summation);

        for (int i = 3 ; i < arraya.length ; i++)
                System.out.println("Square of "+arraya[i]+" = "+(arraya[i]*arraya[i]));
    }
```

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Q4/ Write java program to create 1d array has 5 grads for students , count faild degree
Read items of array from Keyboard using (while)

```
import java.util.Scanner;
public class ArrayExample {
        public static void main(String[] args) {
        Scanner in = new Scanner(System.in);
        int grades[] = new int[5];
        int c = 0; //counter
        System.out.println("write five grade");
        while (c < grades.length){
                grades[c] = in.nextInt();
                 c++; }
        c = 0;
        int failed = 0;
        while (c < grades.length){
        if (grades[c] < 50)
                failed += 1;
                c++; }
        System.out.println("the number of failed : "+failed); }}
```

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Q5/ Write java program to create 1d array has 10 integer numbers then split in to new array A , B each one has 5 items.

The first solution:

```
public class ArrayExample {
        public static void main(String[] args) {
        int [] numbers = {1,2,3,4,5,6,7,8,9,10};
        int [] A = new int[5];
        int [] B = new int[5];

        for (int i = 0 ; i < 5 ; i++)
           A[i] = numbers[i];

        for (int i = 5 ; i < 10 ; i++)
           B[i-5] = numbers[i];

    }
}
```

The second solution:

```
public class ArrayExample {
        public static void main(String[] args) {
        int [] numbers = {1,2,3,4,5,6,7,8,9,10};
        int [] A = new int[5];
        int [] B = new int[5];
        for (int i = 0 ; i < numbers.length; i++) {
          if (i < 5)
              A[i] = numbers[i];
          else if (i >= 5)
              B[i-5] = numbers[i];
        }

    }
}
```

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Q6/ Write Java program to create array have 5 string items {Ali, Hassan , Mohammed, Jaafar , jameel} Then search for word Ali , if his name in array print found ,if not not found.

```
public class ArrayExample {
        public static void main(String[] args) {
        String student[] = {"Ali", "Hassan" , "Mohammed", "Jaafar" , "Jameel"};

        int flag = 0;
        for (int i = 0 ; i < student.length ; i++){
                if (student[i] == "Ali")
                flag = 1;
        }

        if (flag == 1)
            System.out.println("found");
        else
            System.out.println("not found");
    }
```

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Q7/ Write java codes to create array with any five integer items { 5,4,2,1} then sorting array in ascending order.

```
public class ArrayExample {
        public static void main(String[] args) {
        int swap;
        int [] numbers = {5 ,3 , 4 ,1 , 2};
        for (int i = 0 ; i < numbers.length ; i++)
            for (int j = i+1 ; j < numbers.length ; j++){
                if (numbers[j] < numbers[i]){
                swap = numbers[i];
                numbers[i] = numbers[j];
                numbers[j] = swap;
                }
        }

    for (int i = 0 ; i<numbers.length ; i++)
     System.out.println(numbers[i]);
  }
```

5	3	4	1	2

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Q8/ Write java codes to create array with any five integer items {10,20,60,70,30} then find Max and Min Numbers in array

```
public class ArrayExample {
        public static void main(String[] args) {
        int [] numbers = {10,20,60,70,30} ;
        int max = numbers[0];
        int min = numbers[0];
        for (int i = 0 ; i < numbers.length ; i++){
            if (numbers[i] > max)
                max = numbers[i];

            if (numbers[i] < min)
                min = numbers[i];
        }
      System.out.println(max);
      System.out.println(min);
    }
```

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Q1/ Create integer array (1, 2, 2, 3, 3, 3) Count Frequency of Each Element

```
class Main {
   public static void main(String[] args) {
      int[] a = {1,2,2,3,3,3};
      byte[] b = new byte[a.length];
      int numberOfRepeat = 0;
      for (int i=0 ; i<a.length ; i++){
         numberOfRepeat = 0;
         if (b[i] == 1)
            continue;
         for (int j = 0 ; j<a.length ; j++){
            if (a[j] == a[i]){
               numberOfRepeat += 1;
               b[j] = 1; }    }
         System.out.println(a[i]+" Reapeted "+numberOfRepeat + " times");
      }
   }
```

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Q2/ Create integer array (0, 1, 0, 3, 1 ,2) then Shift All Zeroes to the End

```
class Main {
   public static void main(String[] args) {
      int[] a = {0 , 1 , 0 , 3 , 12};
      int temp = 1;
      int temp2 = 0;
      for (int i=0 ; i <a.length-temp ; i++){
         if (a[i] == 0){
            temp2 = a[a.length-temp];
            a[i] = temp2;
            a[a.length-temp] = 0;
            temp += 1; }}
      for (int i=0 ; i <a.length ; i++){
         System.out.println(a[i]);}
      } }
```

0	1	0	3	1	2

↓

2	1	0	3	1	0

↓

2	1	1	3	0	0

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Q3/ Create any two integer array from Keyboard, then Merge Two Sorted Arrays

```
import java.util.Scanner;
class Main {
   public static void main(String[] args) {
      Scanner in = new Scanner(System.in);
      System.out.print("write size of first array: ");
      int a[] = new int[in.nextInt()];
      for (int i=0 ; i<a.length ; i++){
         System.out.print("write value : ");
         a[i] = in.nextInt();}
      int temp = 0;
      for (int i=0 ; i<a.length ; i++){
         for (int j=i+1 ; j<a.length ; j++){
            if (a[j]<a[i]){
               temp = a[i];
               a[i] = a[j];
               a[j] = temp;}
         }
      }
      System.out.print("write size of second array: ");
      int a2[] = new int[in.nextInt()];
      for (int i=0 ; i<a2.length ; i++){
         System.out.print("write value : ");
         a2[i] = in.nextInt();}
      for (int i=0 ; i<a2.length ; i++){
         for (int j=i+1 ; j<a2.length ; j++){
            if (a2[j]<a2[i]){
               temp = a2[i];
               a2[i] = a2[j];
               a2[j] = temp; }} }
      int newArray[] = new int[a.length+a2.length];
      for (int i=0 ; i<a.length+a2.length ; i++){
         if (i<a.length){ newArray[i] = a[i];}
         else{newArray[i] = a2[i-a.length]; }}
      for (int i=0 ; i<newArray.length ; i++){
         for (int j=i+1 ; j<newArray.length ; j++){
            if (newArray[j]<newArray[i]){
               temp = newArray[i];
               newArray[i] = newArray[j];
               newArray[j] = temp; }} }
      for (int i=0 ; i<newArray.length ; i++){
         System.out.println(newArray[i]); }
      }}
```

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Q4/ Write java program to create 1d array has 5 grads for students , count faild degree
Read items of array from Keyboard using (while)

```
import java.util.Scanner;
public class ArrayExample {
        public static void main(String[] args) {
        Scanner in = new Scanner(System.in);
        int grades[] = new int[5];
        int c = 0; //counter
        System.out.println("write five grade");
        while (c < grades.length){
                grades[c] = in.nextInt();
                 c++; }
        c = 0;
        int failed = 0;
        while (c < grades.length){
        if (grades[c] < 50)
                failed += 1;
                c++; }
        System.out.println("the number of failed : "+failed); }}
```

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Q5/ Create array from Key board then Sum of All Elements

```
import java.util.Scanner;
class Main {
   public static void main(String[] args) {
      Scanner in = new Scanner(System.in);
      System.out.println("write number of elements : ");
      int a[] = new int[in.nextInt()];
      System.out.println("write values : ");
      int sum = 0;
      for (int i=0 ; i<a.length ; i++){
         a[i] = in.nextInt();
         sum = sum + a[i];
      }
      System.out.print("summation "+sum);

   }
```

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Q6/ Create array (10, 20, 4, 45, 99) then, Find Second Largest Element Without Sorting

```
class Main {
   public static void main(String[] args) {
      int [] a = {10,20,4,45 , 99};
      int max = a[0];
      int max2 = a[0];
      for (int i=0 ; i<a.length ; i++){
         if (a[i] > max)
            max = a[i];}
      for (int i=0 ; i<a.length ; i++){
         if (a[i] == max){    }
         else if (a[i] > max2){
            max2 = a[i];}
      }
      System.out.print(max2);  }
```

10	20	4	45	99

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Q7/ Write java codes to create array with any five integer items { 5,4,2,1} then sorting array in ascending order.

```
public class ArrayExample {
        public static void main(String[] args) {
        int swap;
        int [] numbers = {5 ,3 , 4 ,1 , 2};
        for (int i = 0 ; i < numbers.length ; i++)
            for (int j = i+1 ; j < numbers.length ; j++){
                if (numbers[j] < numbers[i]){
                swap = numbers[i];
                numbers[i] = numbers[j];
                numbers[j] = swap;
                }
        }

    for (int i = 0 ; i<numbers.length ; i++)
     System.out.println(numbers[i]);
  }
```

5	3	4	1	2

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Q8/ Create array then rotate Array to Left by Given Positions =2

```
import java.util.Scanner;
class Main {
   public static void main(String[] args) {
      int a[] = {1,2,3,4,5};
      int temp;
      for (int i=0 ; i<2 ; i++){
         temp = a[0];
         for (int j=0 ; j<a.length-1 ; j++){
            a[j] = a[j+1];
         }
         a[a.length-1] = temp;
      }

      for (int i=0 ; i<5 ; i++){
         System.out.println(a[i]); }
   }
```

1	2	3	4	5

2	3	4	5	1

3	4	5	1	2

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Q9/ Create array from the keyboard then print the array from second position after that Find Missing number in Consecutive Integers

```
import java.util.Scanner;
class Main {
   public static void main(String[] args) {
      Scanner in = new Scanner(System.in);
      System.out.print("write size : ");
      int a[] = new int[in.nextInt()];
      for (int i=0 ; i<a.length ; i++){
         System.out.print("write values : ");
         a[i] = in.nextInt();
      }

      for (int i=1 ; i<a.length ; i++){
            System.out.println(a[i]);}
```

```
int x = 0 ;
for (int j=a[0] ; j<=a[a.length-1] ; j++){
        if (a[x] == j){
           x+=1;}else{
           System.out.println(j);} }
}}
```

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Q10/ create array from keyboard then Calculate Average Value

```
import java.util.Scanner;
class Main {
   public static void main(String[] args) {
      Scanner in = new Scanner(System.in);
      System.out.print("write size : ");
      int a[] = new int[in.nextInt()];
      int summation = 0;
      for (int i=0 ; i<a.length ; i++){
         System.out.print("write values : ");
         a[i] = in.nextInt();
         summation = summation + a[i];
      }

      System.out.print("summation is : "+summation/a.length);

   }
```

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